SQL

REVIEWER HUB

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1. Creating & Managing Databases and Tables

CREATE DATABASE

DEFINITION
Creates a brand-new, empty database on the server. Everything else (tables, data) lives inside it.

BASIC EXAMPLE

```
CREATE DATABASE D1_Profile;
```

USE

DEFINITION
Switches your active/working database, so the commands that follow run inside that specific database instead of whichever one was open before.

BASIC EXAMPLE

```
USE D1_Profile;
```

CREATE TABLE

DEFINITION
Builds a new table inside the current database by listing its column names and data types.

BASIC EXAMPLE

```
CREATE TABLE D1_Information (
StID INT,
  LName VARCHAR(15),
BDate DATE );
```

DROP TABLE

DEFINITION
Permanently deletes a table and all the data inside it. This cannot be undone.

BASIC EXAMPLE

```
DROP TABLE D1_Information;
```

DROP DATABASE

DEFINITION
Permanently deletes an entire database, including every table inside it.

BASIC EXAMPLE

```
DROP DATABASE D1_Profile;
```

sp_help

DEFINITION

A built-in stored procedure that displays a table's structure — its columns, data types, sizes, and constraints. Great for double-checking a table right after creating or editing it.

BASIC EXAMPLE

```
EXEC sp_help D1_Information;
```

sp_rename

DEFINITION

A stored procedure used to rename an existing object, such as a column, without having to recreate the whole table.

BASIC EXAMPLE

```
sp_rename 'EMPLOYEES.EMP_NO', 'Student_ID';
```

Format: sp_rename 'TableName.OldColumnName', 'NewColumnName'

ALTER TABLE ... ADD

DEFINITION

Adds a brand-new column to a table that already exists.

BASIC EXAMPLE

```
ALTER TABLE EMPLOYEES ADD Total MONEY;
```

ALTER TABLE ... DROP COLUMN

DEFINITION

Removes an existing column (and its data) from a table.

BASIC EXAMPLE

```
ALTER TABLE EMPLOYEES DROP COLUMN CURRENTDATE;
```

ALTER TABLE ... ALTER COLUMN

DEFINITION

Changes the data type or size of a column that already exists.

BASIC EXAMPLE

```
ALTER TABLE EMPLOYEES
ALTER COLUMN DEPARTMENT CHAR(11);
```

PRIMARY KEY / CONSTRAINT

DEFINITION

A PRIMARY KEY is a rule that makes a column's values unique and identifies each row. It is added using a CONSTRAINT, which is simply a named rule attached to a table.

BASIC EXAMPLE

```
ALTER TABLE EMPLOYEES
ADD CONSTRAINT PK_EMPLOYEES PRIMARY KEY (Student_ID);
```

2. Common Data Types

A data type tells SQL Server what kind of value a column is allowed to store.

INT

DEFINITION

Whole numbers (no decimals). Used for IDs, quantities, counts.

BASIC EXAMPLE

```
StID INT
```

SMALLINT

DEFINITION

A smaller-range whole number; takes up less storage than INT.

BASIC EXAMPLE

```
NO_OF_DAYS SMALLINT
```

VARCHAR(n)

DEFINITION

Text of *variable* length, up to a maximum of n characters. Most common type for names, addresses, etc.

BASIC EXAMPLE

```
FName VARCHAR(20)
```

CHAR(n)

DEFINITION

Text of a *fixed* length of exactly n characters (padded with spaces if shorter). Good for short, consistent codes like initials.

BASIC EXAMPLE

```
Minitial CHAR(2)
```

MONEY

DEFINITION

Currency values with decimals, built specifically for storing money amounts.

BASIC EXAMPLE

```
Allowance MONEY
```

DATE

DEFINITION

Calendar dates only (year, month, day) — written as 'YYYY-MM-DD'.

BASIC EXAMPLE

```
BDate DATE
```

3. NULL and NOT NULL

NULL

DEFINITION

Means a value is unknown, empty, or missing for that cell — not the same as zero or blank text.

BASIC EXAMPLE

```
INSERT INTO EMPLOYEES (EMP_NO, Salary) VALUES (300011, NULL);
```

NOT NULL

DEFINITION

A rule placed on a column when creating a table, forcing every row to have a real value in that column — it can never be left empty.

BASIC EXAMPLE

```
EMP_NO INT NOT NULL
```

IS NULL / IS NOT NULL

DEFINITION

Used inside a WHERE clause to filter rows based on whether a column has a value or not. You cannot use '= NULL'; you must use IS NULL.

BASIC EXAMPLE

```
SELECT * FROM Stocks_D1 WHERE Price IS NULL;
SELECT * FROM Stocks_D1 WHERE Price IS NOT NULL;
```

4. Adding, Viewing, Changing & Removing Data

INSERT INTO ... VALUES

DEFINITION

Adds one or more new rows of data into a table.

BASIC EXAMPLE

```
INSERT INTO D1_Information
(StID, LName, FName, Allowance) VALUES (50001271, 'MANZANO', 'RICARDO',
100);
```

SELECT

DEFINITION

Retrieves (reads) data from a table so you can view it. Listing specific column names shows only those columns.

BASIC EXAMPLE

```
SELECT LName, FName FROM D1_Information;
```

SELECT * (asterisk)

DEFINITION

The asterisk is a shortcut meaning 'all columns.' It displays every column in the table.

BASIC EXAMPLE

```
SELECT * FROM D1_Information;
```

WHERE

DEFINITION

Filters which rows are affected or shown, based on a condition. Used with SELECT, UPDATE, and DELETE.

BASIC EXAMPLE

```
SELECT * FROM D1_Information
WHERE Allowance <= 500;
```

Comparison Operators (=, <=, >=, <, >, <>)

DEFINITION

Used inside WHERE to compare a column's value to something: equal to, less/greater than (or equal to), or not equal to (<>).

BASIC EXAMPLE

```
SELECT * FROM D1_Information WHERE Allowance <= 700;
```

AND / OR

DEFINITION

Combine two or more conditions in a WHERE clause. AND requires every condition to be true; OR requires only one of them to be true.

BASIC EXAMPLE

```
... WHERE Allowance <= 500 AND BDate >= '2007-08-13'; ...
WHERE Allowance <= 500 OR BDate >= '2007-08-13';
```

LIKE and the % Wildcard

DEFINITION

LIKE searches for a text pattern instead of an exact match. The % symbol stands for 'any characters, any length,' so '%Rizal%' matches anything containing the word Rizal.

BASIC EXAMPLE

```
SELECT * FROM D1_Information
WHERE Address LIKE '%Rizal%';
```

UPDATE ... SET ... WHERE

DEFINITION

Changes existing values in a table. SET names the column(s) to change and their new value; WHERE limits the update to specific rows (skipping WHERE would update every row!).

BASIC EXAMPLE

```
UPDATE D1_Information SET Address = 'Valley Gulf, Rizal'
WHERE Allowance <= 500;
```

Arithmetic in UPDATE/SELECT

DEFINITION

Columns can be calculated using math operators (*, /, +, -) to compute a new value, such as multiplying rate by number of days to get a salary.

BASIC EXAMPLE

```
UPDATE EMPLOYEES
SET Salary = RATEPERDAY * NO_OF_WORK;
```

DELETE FROM ... WHERE

DEFINITION

Removes entire rows of data that match a condition. Without WHERE, it deletes every row in the table.

BASIC EXAMPLE

```
DELETE FROM D1_Information
WHERE Address LIKE '%Rizal%';
```

5. Comments

-- (Single-Line Comment)

DEFINITION

Anything after two dashes on a line is ignored by SQL Server. Used to leave notes or explanations for yourself and others reading the code — it does not run as a command.

BASIC EXAMPLE

```
-- This inserts a new employee record
INSERT INTO EMPLOYEES VALUES (300011, 'MARTIN', 'JOHN');
```

Tip: Practice typing each example yourself in SSMS. Try changing the table/column names and values to see how the results change — that repetition is what makes these commands stick!